

Education

- 2023 – Present **Candidate, Ph.D. Physics**, University of Michigan, Ann Arbor, MI
Fall
Advisors: Prof. Igor Jovanovic, Prof. James Wells
2025 M.S. Physics
Winter
- 2020 – 2023 **B.S. Physics, Summa Cum Laude**, Florida State University, Tallahassee, FL
Fall Spring
Honors Thesis: Study of Displaced Vertex Tagging with the CMS Experiment
Advisor: Prof. Ted Kolberg
- 2018 – 2020 **Foundational Coursework**, Emory University, Atlanta, GA
Fall Spring

Publications

4. **E. Todd**, V. Fondement, and I. Jovanovic, "Particle Identification in YAP:Ce," *under review*, 2025
3. C. Graham, **E. Todd**, C. Wilson, et al., "Light Output-Dependent Integration Gating for Optimized Pulse Shape Discrimination," *under review*, 2025
2. **CMS HGCAL Collaboration**, "Leakage current of high-fluence neutron-irradiated 8" silicon sensors for the CMS Endcap Calorimeter Upgrade," *under review*, 2025. [Online]. Available: <https://arxiv.org/abs/2510.17280>
1. K. Yocum, H. Smith, **E. Todd**, et al., "Millimeter/Submillimeter Spectroscopic Detection of Desorbed Ices: A New Technique in Laboratory Astrochemistry," *The Journal of Physical Chemistry A*, vol. 123, no. 40, pp. 8702–8708, 2019, ISSN: 10.1021/acs.jpca.9b04587

Research Experience

- 2024 – Present **Graduate Research Assistant**, University of Michigan, Ann Arbor, MI
Summer
Advisor: Prof. Igor Jovanovic
- Developed machine learning (ML) based particle identification techniques for inorganic scintillators in which traditional techniques, *i.e.* charge comparison, are ineffective.
 - Simulated detection efficiency of gamma rays emitted following thermal neutron capture in lithium and germanium samples in preparation for experiments probing low energy nuclear recoil physics via gamma-ray tagging.
- 2022 – 2023 **Laboratory Assistant**, Florida State University, Tallahassee, FL
Summer Summer
Advisor: Prof. Rachel Yohay
- Conducted process quality control (PQC) testing for the CMS High Granularity Calorimeter (HGCAL) to monitor the quality/stability of the silicon sensor production process.
 - Analyzed and presented PQC results to the HGCAL sensor testing group.
 - Helped create and implement a standardized testing procedure at FSU.
- 2021 – 2023 **Undergraduate Research Assistant**, Florida State University, Tallahassee, FL
Fall Spring
Advisor: Prof. Ted Kolberg
- Participated in a deep neural network ML based search for beyond the Standard Model long-lived particles using the CMS Experiment.
 - Trained ML models on Monte Carlo data with varying signal (mass, lifetime) and background (t , b physics) to verify their ability to identify displaced vertices independently of the particular physics.
 - Investigated the most effective parameters of ML models; verified physics-based expectations.
- 2019 – 2019 **Summer Undergraduate Research Experience Fellow**, Emory University, Atlanta, GA
Summer Fall
Advisor: Prof. Susanna Widicus Weaver/Dr. Katarina Yocum
- Conducted proof-of-concept millimeter/submillimeter spectroscopic measurements of H₂O and D₂O binding energy, sublimation enthalpy, and sublimation entropy.
 - Assisted in data collection, analysis, and literature review.

2018 – 2020
Fall Fall

Undergraduate Research Assistant, Emory University, Atlanta, GA

Advisor: Prof. Susanna Widicus Weaver

- Performed millimeter/submillimeter spectroscopic measurements of desorbed astrophysical ice analogs.
- Analyzed broadband molecular line surveys using the GOBASIC software package.

Teaching Experience

2025 – 2026
Fall Winter

Lead Graduate Student Instructor, University of Michigan, Ann Arbor, MI

- Oversee 30 graduate student instructors teaching general physics laboratories with a total enrollment of 1800 undergraduate students.
- Edit and maintain the laboratory manuals and Jupyter notebook worksheets used in the course.
- Manage the course information and organization via Canvas.
- Arbitrate student and graduate student instructor concerns.

2023 – 2025
Fall Spring

Graduate Student Instructor, University of Michigan, Ann Arbor, MI

- 2025 Physics 251, Fundamental Physics for the Life Sciences II Lab
Spring
- 2024 Physics 121, Physics for Architects Lab
Fall
- 2023 – 2024 Physics 241, General Physics II Lab
Fall Winter

2022
Spring

Physics Learning Assistant, Florida State University, Tallahassee, FL

PHY 2049C, General Physics B

2019
Fall

Laboratory Teaching Assistant, Emory University, Atlanta, GA

CHEM 150L, Structure and Properties Lab

Honors, Awards, and Grants

2025
Fall

Rackham Graduate Student Research Grant

Awarded to conduct research at the Paul Scherrer Institute on the ionization yield of high-purity germanium detectors.

2023
Spring

Phi Beta Kappa Society

2023
Spring

Joseph Lanutti Undergraduate Research Award

Awarded for third place in the FSU physics departmental poster presentation contest.

2022
Fall

Evelyn and John Baugh Research Presentation Scholarship

Awarded to travel and present research at SESAPS.

2022
Spring

Anna Runyan Undergraduate Endowment

Awarded for outstanding academic success in physics.

Computing

Languages

Python:

- NumPy
- SciPy
- PyROOT/uproot
- Scikit-Learn

C++:

- ROOT
- GEANT4

Nix:

- Custom nixpkgs
- NixOS configuration

L^AT_EX:

- Self-explanatory...
- Beamer

Operating Systems

Linux (NixOS, Debian/Ubuntu, Arch Linux), MacOS

Skills

Programming	C++: Python: C: (modify and build Chromium EC firmware and coreboot/edk2 firmware).	ROOT, NumPy, SciPy, Scikit-Learn, PyROOT/uproot	GEANT4
Software	GEANT4 (Monte Carlo radiation transport), ROOT (data analysis), Nix/NixOS (package management and system configuration), $\text{\LaTeX}$ (including Beamer).		
Hardware	Use of CAEN digitizers and associated software (CoMPASS, wavedump) for data acquisition; soldering and electronics assembly for experimental setups.		
Operating Systems	Linux (NixOS, Debian/Ubuntu, Arch Linux), macOS.		

Strong preference for open-source software and hardware.

Presentations

4. **E. Todd**, V. Fondement, and I. Jovanovic, "Particle Identification in Fast Inorganic Scintillators via Machine Learning," *Fall Meeting of the Eastern Great Lakes Section of the American Physical Society, Ypsilanti, MI*, 2025
3. **E. Todd** and I. Jovanovic, "Investigation of Anomalously High Ionization Produced by Low Energy Nuclear Recoils in Germanium (poster)," *DOE NNSA Office of Defense Nuclear Nonproliferation R&D University Program Review, Gainesville, FL*, 2025
2. **E. Todd**, A. A. Kadhim, N. Bower, et al., "Process Quality Control for HGCAL at Florida State University (poster)," *89th Annual Meeting of the Southeastern Section of the American Physical Society, Oxford, MS*, 2022
1. **E. Todd**, K. Yocum, P. Gerakines, et al., "A Novel Use of Rotational Spectroscopy for Studying Characteristics of Desorbed Interstellar Ice Analogs (poster)," *Emory SURE 2019 Symposium, Atlanta, GA*, 2019